

# An Investigation of Degree Reduplication in Mandarin

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# Degree Word Doubling



- In Mandarin, doubling a degree word strengthens the meaning.

(1) John fēicháng xǐhuān tā.  
very like 3SG.

‘John likes it very much.’

[e.g. LIKE-DEGREE = 0.6]

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[e.g. LIKE-DEGREE = 0.6]

(2) John **fēicháng fēicháng** xǐhuān tā.  
very very like 3SG.

‘John likes it **very very much**.’

[e.g. LIKE-DEGREE = 0.8]

# Degree Reduplication Asymmetry



■ **Puzzle** (observed but unexplained in ZHAO 2008):

(3) 'John likes it **very very** much.'

a. John **fēicháng fēicháng** xǐhuān tā.  
very very like 3SG

b. \*John **hěn hěn** xǐhuān tā.  
very very like 3SG

# Degree Reduplication Asymmetry



- Disyllabic degree words can be doubled:  
*fēicháng, chāojí, tèbié, etc.*
- **Monosyllabic** degree words **CANNOT** be doubled:  
*hěn, chāo, tè, etc.*
- By 'degree word', we refer to elements that fill in the degree-argument of gradable predicates, excluding composite degree-related expressions like *yǒudiǎn* 'a bit' and the superlative particle *zuì*.

# Outline



- 1 More Empirical Data
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- 3 More on the Constraint  $RED_{XP} \geq F_T$
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## Observations:

- It is misleading to say that monosyllabic degree words cannot be doubled;
- Monosyllabic words can actually appear in reduplication.

# Monosyll Deg Wrds Actually CAN be Redup-ed

- Doubling *hěnn* alone is ungrammatical;
- It becomes grammatical once we reduplicate along more materials.

(4) John [hěnn \*(xǐhuān (tā))] <sub>ℝ</sub> [hěnn xǐhuān tā] <sub>ℝ</sub>.  
very like 3SG very like 3SG  
'John likes it very very much.'  
✗[hěnn] <sub>ℝ</sub> but ✓[hěnn xǐhuān] <sub>ℝ</sub> or ✓[hěnn xǐhuān tā] <sub>ℝ</sub>

# Disyllabic Deg Words

- Disyllabic degree words can also have more reduplicated materials along.

(5) John [fēicháng (xǐhuān (tā))]<sub>R</sub> [fēicháng xǐhuān tā]<sub>B</sub>.  
very like 3SG very like 3SG  
'John likes it very very much.'  
✓[fēicháng]<sub>R</sub> or ✓[fēicháng xǐhuān]<sub>R</sub> or ✓[fēicháng xǐhuān tā]<sub>R</sub>

# The Reduplication is Phrasal

- We conclude that the reduplication here is phrasal;
- Some factors force monosyllabic degree words to not appear alone in the reduplicant;
- We will articulate what these factors are in the following sections.

(6) John RED- [hěnfēicháng xǐhuān tā]<sub>B</sub>.  
                  very                  like          3SG  
          ‘John likes it very very much.’

# Total Reduplication of the Phrase?

- Total reduplication gets deviant as the phrase becomes logner.

(7) ?? John[hěn xǐhuānhē hēi kāfēi]<sub>R</sub>[hěn xǐhuānhē hēi kāfēi]<sub>B</sub>.  
verylike drinkblackcoffee verylike drinkblackcoffee  
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- But it does not sound categorically bad, and there is also speaker and register variation;
- Under a constraint-theoretic framework, it can be captured by placing **MAX-BR** somewhere *ad hoc* inside the sequence

**RED ≤ σσσσσσ** >> **RED ≤ σσσσσσ** >> ... >> **RED ≤ σσσ** >> **RED ≤ σσ**

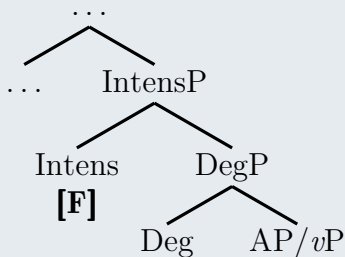
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# Reduplication of DegP

- We adopt the syntactic structure in (8);

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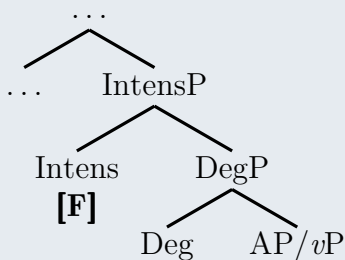


- the gradable predicate (AP or  $vP$ ) is headed by  $Deg^\circ$  (see ABNEY 1987, KENNEDY 1999 *i.a.*);
- $Intens^\circ$  (same as  $F_{RE}^\circ$  in WANG 2023) merges with DegP;

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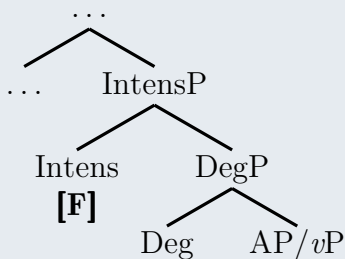
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(10) The critical constraint:

$$\text{RED}_{\text{XP}} \geq \text{F}_T$$

Phrasal reduplicants are minimally one foot (two syllables).

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- **[F]**  $\leftrightarrow$  RED is subject to a minimality constraint (10).

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- It is not morpheme-specific;
- No phrasal reduplicants can underlength a foot (11);

(11) reduplication of  $vP \rightsquigarrow$  iterative reading  
Zhěngtiān {**dǎyóuxì** dǎyóuxì/ **xīxì** xīxì/ \***wán** wán}.  
day.long play.game play.game play play play play  
'He is {playing video games/ playing/ playing} all day long.'

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- (12) Zhěngtiān [{wán wán/ wán ya}]<sub>R</sub> [<sub>vP</sub>wán]<sub>B</sub>.  
day.long play play play PRT play
- (13)  $\text{RED}_{\text{XP}} \geq \text{FT} \gg \{\text{DEP-BR}, *\sigma\}$

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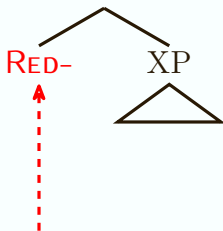
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- Notice that the phrasal base in (12) is itself monosyllabic;
- Morpheme- (or word-) level reduplicants (see LEE-KIM 2016 for

# A Visualisation of the Constraint

- **ALL phrasal reduplicants are no less than one foot.**



☹ if realised as less than one foot

## FTBIN is Not Enough

- Foot binarity is **NOT** enough to cover the pattern;
- Notice that (14a) passes the foot binarity filter;
- There is also no need for the base to left align with a foot;

(14) a. \*zhěngtiān (wán wán)<sub>FT</sub>  
day.long play play  
b. zhěngtiān (**wán wán**)<sub>FT</sub> (**wán wán**)<sub>FT</sub>  
day.long play play play play

- However, foot binarity may still play a role in the end. For example, it seems that a zero syllable (DUANMU 1999) is accommodated to the base in (15) (see also HUANG 2017).

(15) zhěngtiān (**wán wán**)<sub>FT</sub> (wán  $\emptyset_\sigma$ )<sub>FT</sub>  
day.long play play play

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# Implications

- **All phrasal reduplicants** in Mandarin are subject to **the same minimality requirement**;
- In reduplication, phonology must know the **syntactic structure** of the base *i.e.* whether it is an XP.

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# Multiple Intens<sup>o</sup>'s?

- Expectedly, for monosyllabic degree words, repeating itself several times is also an option (16);

- (16) a. \*John hěn hěn gāo.  
          very very tall
- b. John hěn hěn hěn gāo.  
          very very very tall
- c. John hěn hěn hěn hěn gāo.  
          very very very very tall

- One problem is whether we need to encode the difference between (16b) and (16c) in syntax, or attribute it to how RED- is implemented in phonology.

# DegP-External Particles Never Appear in Red

- Certain degree words co-occur with particular sentence final particles (see SOH 2009, XIONG & HSIEH 2021);
- These SFP's may clitise into the base, but never into the reduplicant (17).

- (17) a. Zhè dào wèntí [tài nán -le]<sub>ω</sub>!  
this CLF problem too difficult -PFV  
'This problem is too difficult.'
- b. Zhè dào wèntí [tài nán (\*-le)]<sub>ω</sub> [tài nán -le]<sub>ω</sub>!  
this CLF problem too difficult -PFV too difficult -PFV  
'The problem is too too difficult!'

# An Example of Both Head & Phrasal Redup

- Deg<sup>o</sup> itself can be realised as RED-;

(18) '(?) somewhat beautiful'

- a. měi~měi-de  
beautiful~DEG-DE
- b. piào<~piào>liàng<~liàng>-de  
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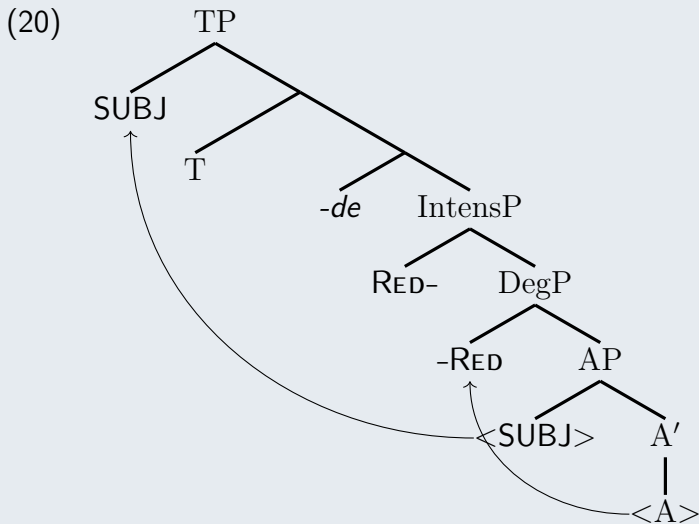
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- Intens<sup>o</sup> can be used together with (18).

- (19) a. měi~měi-                  měi~měi-de  
beautiful~DEG- beautiful~DEG-DE
- b. piào<~piào>liàng<~liàng>-    piào<~piào>liàng<~liàng>-de  
beautiful<~DEG>-                  beautiful<~DEG>-DE

# An Example of Both Head & Phrasal Redup



# THANK YOU

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